EXPERIMENT 23

Implement the top-hat morphological operation using OpenCV.

Aim

Implement the top-hat morphological operation using OpenCV.

Algorithm

• Import OpenCV and NumPy.

• Read the input image.

• Convert the image to grayscale.

• Apply the top-hat operation with a structuring kernel.

• Save the top-hat image.

Code

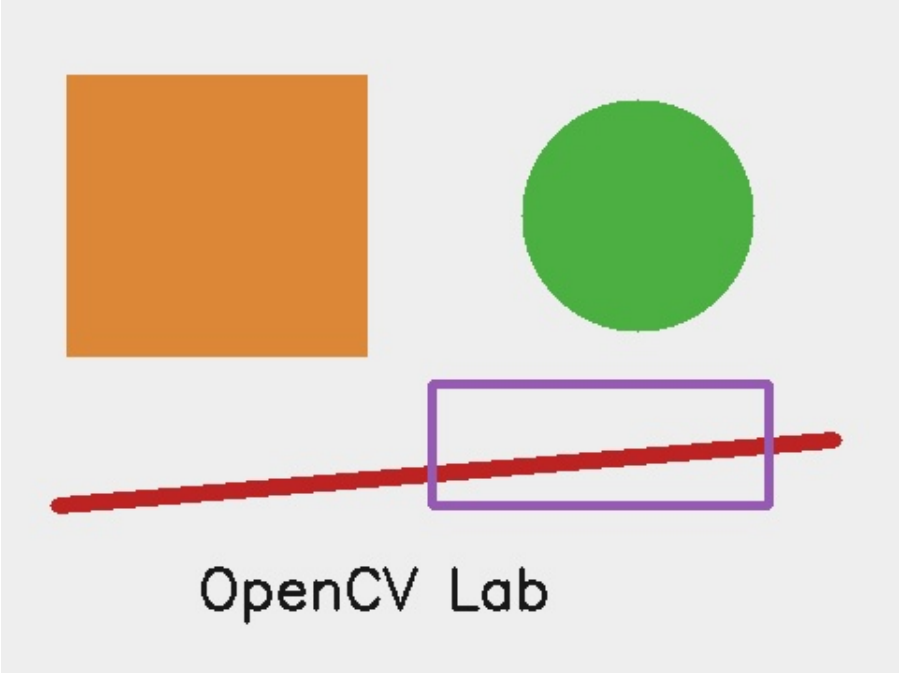
```
import cv2
import numpy as np

img = cv2.imread("images/input.jpg")
if img is None:
    raise FileNotFoundError("images/input.jpg not found")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
kernel = np.ones((5, 5), np.uint8)
result = cv2.morphologyEx(gray, cv2.MORPH_TOPHAT, kernel)
cv2.imwrite("output_images/experiment23_top_hat.png", result)
print("Saved output_images/experiment23_top_hat.png")
```

Input Image:

Output Image:

Input Image



Output Image